

**Design and manufacturing services.** For telecom products design and manufacturing services, apart from achieving increased complexity of designs at lower costs, lesser time to market is a critical factor. This demand is met by many firms using concurrent engineering where different tasks for developing a system are performed in parallel.

**Infrastructure development.** With an increased number of affordable devices and increased adoption of latest technologies, there is a major push from customer end for better connectivity. This thrust to connect remote locations positively affects telecom infrastructure developers. "Improvements in technology also help us serve more customer segments," says Dhaval Bhagora, marketing manager - telecom solutions, Matrix Comsec.

Business organisations are now looking for integrated communication solutions to improve their services. Connectivity within offices, between branch offices, and in sales and supply chains are some new trends as seen by Matrix. "With growing volumes, components and products also become cheaper," adds Bhagora. This is a good phenomenon for manufacturers and infrastructure providers. The challenge here is to install systems that have low maintenance and are easily upgradable.

## Business opportunities across several verticals

Fifth-generation (5G) networks would use an expensive communication technology, predict industry experts, and would prove more beneficial for business sector rather than individual consumers. "This is due to the ability of LTE over its predecessors to prioritise the customer or service grade," comments Srinivasa Moorthy, director of design engineering, D4X Technologies Pvt Ltd.

Moreover, LTE offers better mobility, speed, security, bandwidth

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and ease of setting up as compared to Wi-Fi, making it suitable for business models involving the Internet of Things (IoT) and machine-to-machine (M2M) communication.

An article published by Arthur D. Little titled 'The Business Benefits of 4G LTE' suggests that enterprise mobility of 4G and advanced technologies can bring increased sales, improved customer service, reduction in cost, flexibility, agility and decision-making capabilities in businesses.

**Healthcare.** We find that wellness and health industry is very enthusiastic about developing telecom technologies, and it is expected to be one of the most benefited sectors.

Tele-medicine, by virtue of which a doctor at a remote location can treat a patient using video-telephony, will become easier with better streaming and high-quality imaging.

4G can also be used for connecting ambulances to hospitals for in-transit treatment. Such ambulance systems are already employed in University Hospital of Saarland, Germany (Mobile Stroke Unit) and London's Air Ambulance; paramedics can perform CT scans on patients en route and results can be sent to doctors in the emergency room. This has proven to increase the survival rate of stroke patients.

Popularity of 4G is also bringing more focus on preventive medicine and assisted living for the elderly.

A completely fit person can also keep track of his or her health and balance the lifestyle accordingly.

There are several wearable devices and mobile apps in the market or in development stage that can keep record of activities and analyse information on cloud to predict possibility of illnesses.

**Automotive.** With the push for in-car services by consumers and immense R&D going on in the connected car segment by car manufacturers like BMW and Mercedes-Benz, the automotive industry could benefit immensely from the high-speed LTE network and its successors.

Remote condition monitoring, in-vehicle Wi-Fi hotspots, predictive engine diagnostics, real-time navigation and location based services are the notable trends in this segment.

Mahindra Reva, Indian car manufacturer, currently has commercial electric cars that can be controlled by a smartphone app. The company is also undertaking extensive R&D to manufacture driverless cars.

## Definitely a fast growing sector

The Indian telecommunication industry is the second largest in the world and is growing at a rate of 45 per cent, according to the government's Department of Telecommunications.

All telecom services in India are open to private players, and they now hold more than 85 per cent of the share as against five per cent in 1999. With demand for better-quality services, faster rate of technology adoption and innovations in this segment, the telecom sector opens up immense opportunities for the electronics industry. ●